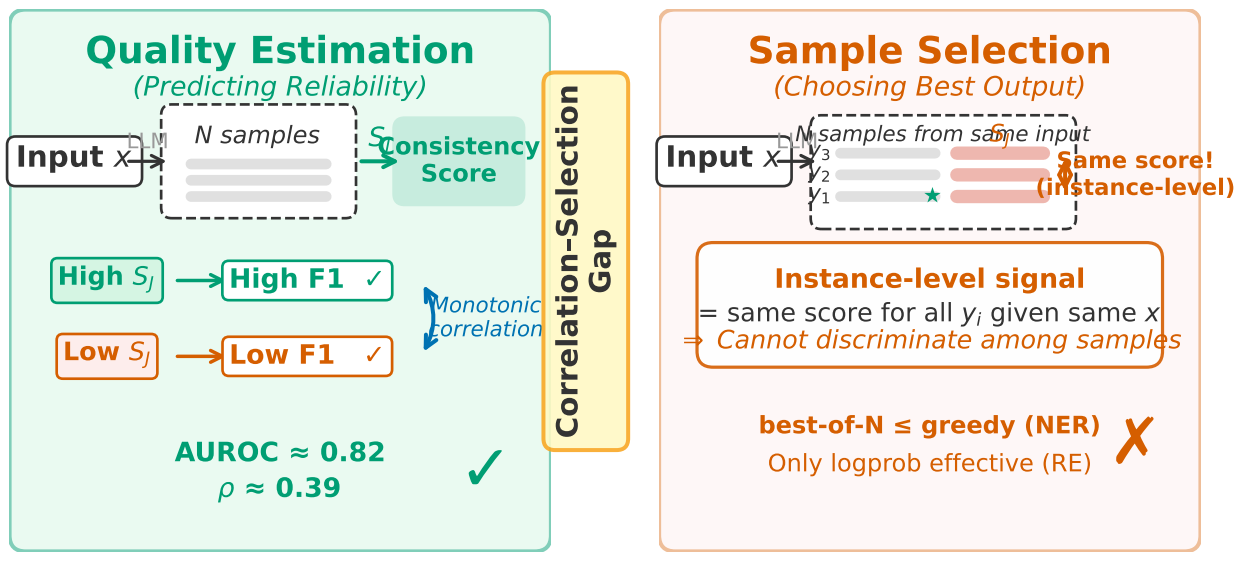
Quality Estimation
(Predicting Reliability)
Input x
LLM
N samples
S_J
Consistency Score
High S_J
High F1 ✓
Low S_J
Low F1 ✓
Monotonic correlation
AUROC ≈ 0.82
ρ ≈ 0.39
✓
Correlation–Selection Gap
Sample Selection
(Choosing Best Output)
Input x
LLM
N samples from same input
S_J
y_3
y_2
y_1
Same score!
(instance-level)
Instance-level signal
= same score for all y_i given same x
⇒ Cannot discriminate among samples
best-of-N ≤ greedy (NER)
Only logprob effective (RE)
✗